

Dina Traykova

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EDUCATION

University of Oxford, Oxford, United Kingdom

Aug 2017 – Present

Doctor of Philosophy (D.Phil) in Astrophysics

- Testing theories of Dark Energy and Modified Gravity on different scales.

University College London, London, United Kingdom

Sep 2013 – Jun 2017

Master of Science (MSci) in Theoretical Physics

- First Class Honours degree.
- Awarded the Tessella Prize for Software - Best use of software in final year (Astro) Physics project

RESEARCH

EXPERIENCE

University of Oxford, Oxford, UK

▪ PhD - Astrophysics, Oxford

Oct 2017– Present

- Project: *Testing Modified Gravity theories on different scales*: in the weak gravity regime using data from Cosmology and in the strong gravity regime using Gravitational waves data from collisions of compact objects.
- Supervisors: Professor Pedro Ferreira and Dr Emilio Bellini
- Research areas: Modified gravity, Parameter estimation, Statistical methods, Numerical relativity, Cosmology.

University College London, London, UK

▪ Masters project - Astrophysics, UCL

Oct 2016 – Mar 2017

- Project: *Accretion of a Scalar Field to a Schwarzschild Black Hole*. In this project I developed the numerical framework for investigating the problem of scalar field accretion to a spherically symmetric black hole by solving the full nonlinear Einstein equations for a scalar field potential numerically.
- Supervisors: Professor Hiranya Peiris and Dr Jonathan Braden
- Research areas: Numerical relativity, cosmology.

University of Oxford, Oxford, UK

▪ Summer internship - Astrophysics, Oxford

Jul 2016 – Aug 2016

- Project: *The Fine-Tuning of the Cosmological Constant*. During this internship I mainly worked on numerically calculating the relative probability of life as a function of time for different cosmological constants. Additionally, I wrote some of the material for an outreach website for the Consolidation of Fine-Tuning project.
- Supervisor: Dr Rafael Alves Batista

University College London, London, UK

▪ Summer placement - High Energy Physics, UCL

Jul 2015 – Aug 2015

- Project: *LAr neutrino detector computer simulations*. The UCL High Energy Physics group has developed a small 10 litre LAr test stand at UCL to test various prototype detector technologies in LAr. My role within the project was to simulate the expected scintillation light levels in the vessel from incoming muons with different energies.
- Supervisor: Dr Anna Holin

AWARDS & SCHOLARSHIPS

- Tessella Prize for Software - University Collage London (2017)
- Oxford Summer Research Programme - University of Oxford (2016)

TEACHING AND OUTREACH

- General Relativity 2017-2018 (3rd year undergraduate practice classes)
- Physics outreach with the Oxford Astrophysics department (2017 - present)

PUBLICATIONS	▪ Carlos García-García, Emilio Bellini, Pedro G. Ferreira, Dina Traykova, and Miguel Zumalacárregui. Theoretical priors in scalar-tensor cosmologies: thawing quintessence. 2019 ▪ S. Joudaki, H. Hildebrandt, D. Traykova, et al. KiDS+VIKING-450 and DES-Y1 combined: Cosmology with cosmic shear. 2019 ▪ Dina Traykova, Emilio Bellini, and Pedro G. Ferreira. The phenomenology of beyond Horndeski gravity. <i>JCAP</i>, 1908:035, 2019 ▪ Dina Traykova, Jonathan Braden, and Hiranya V. Peiris. Accretion of a Symmetry Breaking Scalar Field by a Schwarzschild Black Hole. <i>Phil. Trans. Roy. Soc. Lond.</i>, A376(2114):20170122, 2018		
CODING & HPC EXPERIENCE	▪ Coding languages: Python, Mathematica, C, C++ and Fortran 90. ▪ Codes: • Einstein-Boltzamnn solving codes: Class, hi_class, Camb • Parameter Estimation: CosmoSIS, MontePython, CosmoMC • Numerical Realtivity: SENR/NRPy+, GRChombo. ▪ HPC systems: • Glamdring: Oxford Astrophysics local cluster. Size: $\sim 1,200$ cores • DiRAC @CSD3: Intel Xeon Phi KNL and Intel Skylake Xeon Gold. Cluster size: $\sim 120,000$ cores • SuperMUC-NG: Intel Skylake Xeon Platinum. Cluster size: $\sim 310,000$		
LIST OF TALKS	<table><tr><td> ▪ Boson Star Collapse to a Black Hole in Modified Gravity • Strong Gravity 2019 Workshop, Lisbon, Portugal • European Einstein Toolkit Meeting, London, UK ▪ The Phenomenology of Beyond Horndeski Gravity • 22nd International Conference on General Relativity and Gravitation, Valencia, Spain ▪ The Accelerating Universe - Modified Gravity • Out Thinkers, Pride In STEM, Oxford, UK ▪ Accretion of a Scalar Field to a Schwarzschild Black Hole • Master’s thesis talks, UCL, London, UK ▪ The Apparent Fine-Tuning of the Cosmological Constant • Oxford Summer Research Programme, Oxford, UK </td><td>Academic Talk Oct 2019 Sep 2019 Academic Talk Jul 2019 Public Talk Oct 2018 Academic Talk Mar 2017 Academic Talk Aug 2016</td></tr></table>	▪ Boson Star Collapse to a Black Hole in Modified Gravity • Strong Gravity 2019 Workshop, Lisbon, Portugal • European Einstein Toolkit Meeting, London, UK ▪ The Phenomenology of Beyond Horndeski Gravity • 22nd International Conference on General Relativity and Gravitation, Valencia, Spain ▪ The Accelerating Universe - Modified Gravity • Out Thinkers, Pride In STEM, Oxford, UK ▪ Accretion of a Scalar Field to a Schwarzschild Black Hole • Master’s thesis talks, UCL, London, UK ▪ The Apparent Fine-Tuning of the Cosmological Constant • Oxford Summer Research Programme, Oxford, UK	Academic Talk Oct 2019 Sep 2019 Academic Talk Jul 2019 Public Talk Oct 2018 Academic Talk Mar 2017 Academic Talk Aug 2016
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INTERESTS	▪ Academic interests: Numerical general relativity, gravitational theories, black hole dynamics, scalar fields, dark energy, origin and growth of structure. ▪ Others: Science outreach, rock climbing and mountaineering, cycling, art, martial arts, linguistics.		